

READING PASSAGE 2

The Development of Atomic Theory

A In recent years, the development of atomic theory has attracted considerable attention from both the academic community and the general public. This growing interest reflects a broader recognition of the importance of this topic for understanding some of the most pressing challenges facing contemporary society.

B The origins of contemporary research in this area can be traced to pioneering studies conducted during the mid-twentieth century. These early investigations, though constrained by the technological and methodological limitations of their era, established foundational principles that continue to inform current scholarship. Successive generations of researchers have built upon these foundations, progressively refining theoretical models and expanding the empirical evidence base through increasingly sophisticated research designs.

C At the core of current understanding lies a set of interrelated mechanisms that collectively explain the observed patterns and outcomes. First, structural factors create the underlying conditions within which individual and collective action takes place. These structural dimensions include economic incentives, institutional frameworks, and cultural norms that shape behaviour in ways that are often not immediately apparent to the actors themselves. Second, dynamic feedback processes amplify or dampen initial effects, generating patterns of change that can diverge substantially from what linear models would predict.

D The empirical record is further strengthened by a series of meta-analyses that have synthesised findings from hundreds of individual studies. These comprehensive reviews have consistently identified moderate to large effect sizes, with the most robust findings demonstrating remarkable consistency across different research designs, measurement approaches, and analytical strategies. This convergence of evidence from multiple independent lines of inquiry considerably strengthens confidence in the validity of the core theoretical propositions.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F The future trajectory of research in this area will likely be shaped by several converging trends. Advances in computational methods and data science are opening up new analytical possibilities, enabling researchers to detect patterns and test hypotheses at scales that were previously unimaginable. Simultaneously, growing recognition of the importance of interdisciplinary collaboration is breaking down traditional barriers between fields and fostering innovative syntheses of diverse theoretical perspectives.

G In conclusion, the study of the development of atomic theory represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Questions 1–3

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 1 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.
- 2 The author believes that all research questions in this field have been answered.
- 3 Early researchers had access to the same technologies as modern scholars.

Question 4

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 4 The landmark study analysed data spanning a period of _____ years.

Questions 5–7

Choose the correct letter, A, B, C or D.

- 5 The passage indicates that critics have raised concerns about:
 - A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations
- 6 The meta-analyses cited in the passage reveal that:
 - A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations
- 7 What finding emerged from the landmark study discussed in the passage?
 - A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations

Question 8

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 8 Meta-analyses have confirmed the reliability of the core findings in this field.

Questions 9–10

Choose the correct letter, A, B, C or D.

- 9 The meta-analyses cited in the passage reveal that:
- A. too much data collection
 - B. the generalisability of findings across contexts
 - C. insufficient government funding
 - D. overly complex statistical methods
- 10 What does the passage identify as a significant challenge to current understanding?
- A. excessive funding
 - B. theoretical and methodological limitations
 - C. public interest
 - D. international collaboration

Questions 11–13

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 11 Dynamic _____ processes can amplify or dampen initial effects in complex systems.
- 12 The landmark study analysed data spanning a period of _____ years.
- 13 Case studies have documented the effects of _____ change on the system.